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Technical Report

Transaxillary Totally Endoscopic Robot-Assisted Ansa Cervicalis to Recurrent Laryngeal Nerve Reinnervation for Repair of Unilateral Vocal Fold Paralysis

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Abstract

Pediatric unilateral vocal-fold paralysis represents a source of significant morbidity, for which treatment options are quite limited. Conventional management strategies suitable for adults are not appropriate for the developing larynx. In this study, we report the first experience with minimally invasive laryngeal reinnervation. While open techniques for pediatric recurrent laryngeal nerve reinnervation have been performed, these require large, visible incisions, which limit the appeal of this technique. The transaxillary endoscopic approach to the neck significantly reduces pain and recovery time from cervical surgery. In this study, we report the feasibility of transaxillary totally endoscopic robot-assisted laryngeal reinnervation for unilateral vocal-fold paralysis. Operative time was less than 3 hours, and patients were discharged the day of surgery. No postoperative narcotics were required. Initial results are favorable.

Introduction

Hoarseness related to unilateral vocal-fold paralysis (UVFP) represents a major source of morbidity for children. Treatment options that effectively and durably ameliorate this condition for adults are not suitable for children. Laryngeal framework surgery is inappropriate for the developing larynx and long-term growth effects are not documented. Injection therapies do not provide durable improvement.[1,2] Laryngeal reinnervation has emerged as a validated method for correcting hoarseness related to UVFP in adults. This technique is limited in children because of the need for a large cervical incision.

Minimal access techniques to the cervical compartment have evolved dramatically over the past 10 years. With the development of the transaxillary totally endoscopic approach to the neck, significant procedures, including thyroidectomy, parathyroidectomy, and vagal nerve stimulator placement, can be performed through an endoscopic approach, with a subsequent dramatic reduction in pain and shortening of recovery.[3–7] Some institutions have developed extensive experience with these approaches. Combined with the fine dissection and anastomosis capabilities facilitated through robotic surgery, the capacity to perform laryngeal reinnervation through an endoscopic approach has emerged. In this report, we describe the first application of the transaxillary totally endoscopic approach combined with robotic assistance to perform ansa cervicalis to recurrent laryngeal nerve (RLN) reinnervation. In this paper, we report on operative time, blood loss, complications, and initial results.

Materials and Methods

Pediatric patients undergoing evaluation for hoarseness in our Pediatric Voice Clinic were screened for UVFP. A formal videolaryngostroboscopy was performed. Among those patients with significant hoarseness weighing greater than 50 kg and with UVFP of greater than 2 years in duration, 2 patients were identified and laryngeal reinnervation was offered. Informed consent for the endoscopic robotic-assisted approach was given. All patients had previously failed speech therapy.

Surgical technique

The transaxillary totally endoscopic approach was performed as described by us previously. The patient is placed in the supine position (Fig. 1). Under CO_2 insufflation of 8 mm Hg, using a central 10-mm camera port and two 5-mm instrument ports, dissection was carried across the pectoralis facia from the axilla to the ipsilateral omohyoid muscle. This

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FIG. 1. Patient positioning. The left arm is allowed to be slightly lowered, exposing the midaxillary line.

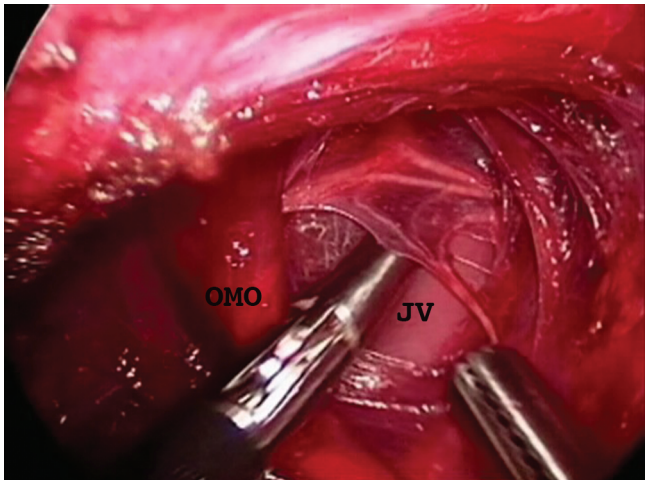


FIG. 2. Endoscopic view of the left carotid sheath. The jugular vein is identified by the right instrument. Omohyoid is retracted inferiorly by the left instrument. The main trunk of the ansa cervicalis is seen passing deep to the omohyoid, running parallel and medial to the jugular vein.

was dissected medially to the region of the jugular vein. The ansa cervicalis was identified anterior to the jugular vein and immediately deep to the omohyoid muscle (Fig. 2). This was dissected distally to its insertion in the sternothyroid muscle. Next, dissection was carried posteriorly to the thyroid gland. The recurrent laryngeal nerve was identified in this region and a second proximally and distally until a full 3 cm was exposed (Fig. 3). At this point, the DaVinci surgical robot system (S-series) was docked (Fig. 4). The RLN was divided approximately 3 cm from its insertion site. The ansa cervicalis was divided approximately 2 cm distal to the omohyoid muscle. Using an 8.0 monofilament suture, anastomosis was performed by using surgical robotics (Fig. 5). Following this, ipsilateral unilateral vocal-fold injection laryngoplasty was performed to provide immediate improvement and facilitate speech-therapy intervention during the period of reinnervation.

Voice assessment

Three subjective voice-quality tools were utilized to assess outcomes by comparing pre- and postoperative results. These tests included the Voice Related Quality of Life Measure (V-RQOL), the Voice Handicap Index (VHI), and the Consensus Auditory Perceptual Evaluation of Voice (CAPE-V). The patient completed all three quality measures within 1 month of surgery and approximately 2 months postoperatively. Speech therapy was commenced both preoperatively and at intervals postoperatively. Postoperative videostroboscopy and voice analysis was performed to document outcomes. Both pre- and postoperative quality-of-life studies and voice-handicap studies were performed.

Results

Two patients underwent transaxillary totally endoscopic laryngeal reinnervation. Both patients had a history of cardiac surgery as newborns resulting in left UVFP. In both cases, the patients had more than a decade of severe hoarseness and had failed speech therapy. At the time of presentation, no additional interventions had been offered for voice rehabilitation.

Operative times were 175 minutes for case 1 and 120 minutes for the second case 2. Less than 5 mL of blood loss occurred. No complications occurred. Both patients were discharged home on the day of surgery without narcotics. Both patients returned to normal activity within 72 hours.

Results of the V-RQOL demonstrated an improvement of 47.5% (47.5 preop to 95% postop) in patient 1 and an improvement of 35% (52.5% preop to 87.5% postop) in patient 2. Results of the VHI demonstrated significant improvement in all three scales, as well as the total scale: physical scale improvement of 26 (32 to 6), functional scale improvement of 8 (13 to 5), emotional scale improvement of 10 (13 to 3), and total score improvement of 44 (58 to 14). Results of the CAPE-V showed postoperative improvement in all categories of overall severity, roughness, breathiness, strain, pitch, and loudness, as determined by the speech pathologist for both patients.

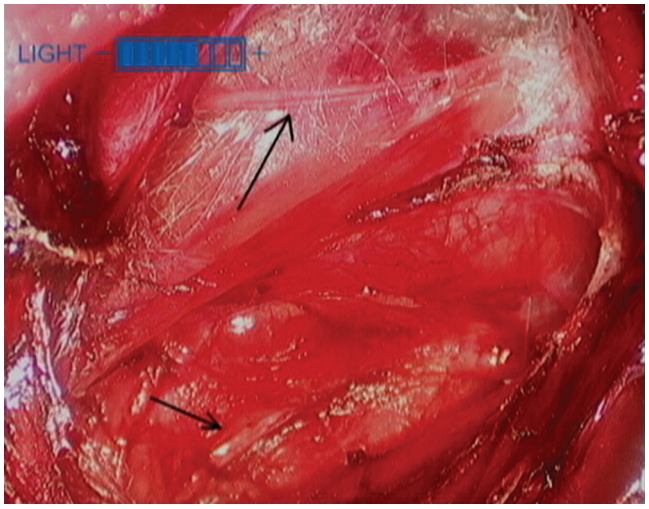


FIG. 3. Endoscopic view demonstrating both the ansa cervicalis (long arrow) and recurrent laryngeal nerve (short arrow) prior to anastamosis.



FIG. 4. Docking of the surgical robot platform to the axilla.

One of the patients was lost to follow-up, and final V-RQOL in the second revealed standard score of 82.5%. VHI showed continued improvement in total score of 9 with preop score of 58. Physical score was 6, functional score 3, and emotional score 0. One-year results of CAPE-V revealed significant improvement in breathiness, roughness, and strain. Pitch continued to be elevated for patient's age and gender, although he reported he was no longer mistaken for a female on the phone.

Discussion

Pediatric UVFP treatment is mainly limited by the inability to perform conventional thyroplasty procedures in the developing larynx. With the validation of RLN reinnervation techniques in the adult population, the possibility of applying this procedure in the pediatric population is compelling. The chief limitation has been the acquisition of a large, visible cervical incision, often in children who have already undergone extensive surgeries. The development of the transaxillary totally endoscopic approach to the neck certainly alters the therapeutic index of this procedure, markedly reducing the morbidity and converting this to a 2–3-hour outpatient endoscopic procedure with virtually no postoperative pain and minimal risk. Further, this does not alter future laryngeal interventions, such as conventional thyroplasty, should the child become a candidate for this, are required this procedure later in life.

In the adult population, quality-of-life studies have demonstrated a significant general health-, and voice-related effect of UVFP. Spector et al. examined quality of life using the SF-36 and two voice-specific instruments, VHI and VOS, before and after thyropolasty. Both general and voice-specific preop quality-of-life instruments demonstrated impairment, compared with normative values; these reversed following thyroplasty. Interestingly, both physical and social functioning domains were reduced preoperatively. Other researchers have demonstrated reduced SF-36 scores among mixed-etiology dysphonic patients. Among three major groups of dysphonia, UVFP had the worst scores. Spector suggests that UVFP affects a patient's self-perception and contributes to feelings of nervousness and depression.

The possibility of adverse psychosocial outcomes of chronic dysphonia in children is self-evident. This relation has been the focus of scientific inquiry since the 1970s. Studies have primary focused on adult perceptions of children with chronic dysphonia, all of which have described a negative perception of children with dysarthric speech.[10] Similarly, adolescent perceptions of children with speech disorders have been studied, with findings of adverse influence related to abnormal speech.[11,12]

Crumley et al. began the clinical application of ansa-cervicalis-to-RLN anastamosis in 1986, and has reported consistently positive results since that time.[13,14] He compared results with injection laryngoplasty, thyroplasty, and ansa-RLN anastaomis. The neurorrhapy compared favorably in several measures, including surgical time, ease of performance, time interval to improvement, reversibility, changes in vocal structure, publications, and cost, in addition to phonotory result.

Many other investigators have demonstrated good to excellent results with this technique. Olsen et al. reported excellent acoustic and perceptual results in a series of 12 patients.[15] In a series of 8 patiens, Zheng et al. reported the return of mucosal waves after the procedure[16] and El-Kashlan et al. had normal voices in each of three patients.[17] May and Beery had excellent results in a series of 20 patients undergoing the ansa-cervicalis reinnervation procedure.[18]

The use of the DaVinci surgical robotics platform has been previously described in combination with the transaxillary totally endoscopic approach to the neck.[6,7] This is an ideal application of this tool, as the excellent optics with stereo vision, in addition to multiple degrees of freedom, allows for precise nerve anastomosis. This report demonstrates an initial experience only, and further study will be necessary to completely characterize the benefits of this surgical approach. Transaxillary totally endoscopic surgery of the neck has been validated in other surgical contexts with no significant increased risk for major complications. The very low morbidity with no apparent loss in anastamosis quality make this an approach promising for voice rehabilitation in the pediatric patient population.

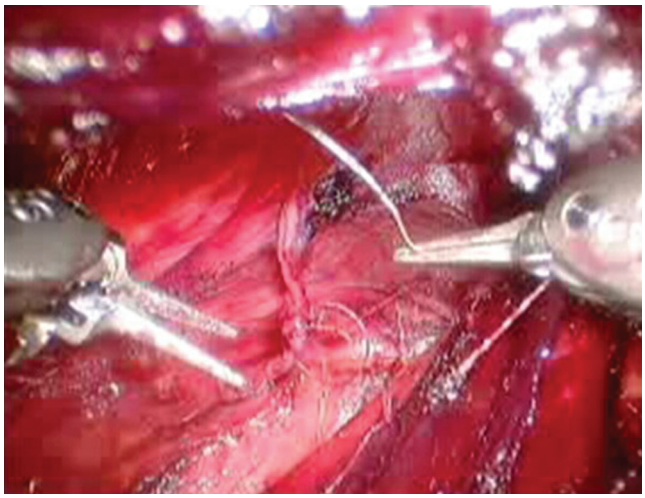


FIG. 5. Docking of the surgical robot platform to the axilla.

Conclusions

Pediatric UVFP is a significant source of long-term morbidity in children; until recently, no durable repair of this condition has been available. With increasing success with laryngeal reinnervation, this option is an obvious choice for voice rehabilitation. The transaxillary totally endoscopic robotic-assisted reinnervation procedure offers precise nerve anastomosis with very low morbidity and can be done as an outpatient procedure with a rapid return to normal activity. The feasibility of this procedure has been established, and further study is warranted.

Disclosure Statement

No competing financial interests exist.

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